Tao Wang

Moscow, Idaho | twang1217.github.io

EDUCATION

Doctor of Philosophy in Computer Science January 2024 – Present
University of Idaho, Moscow, Idaho

Bachelor of Science in Computer Science August 2022 – May 2023
University of Idaho, Moscow, Idaho

RESEARCH EXPERIENCE

Graduate Research Assistant January 2024 – August 2026
Department of Computer Science, University of Idaho, Moscow, Idaho

Directed Research (First Author): "A Temperature-Dependent Geometry-Corrected Stress Descriptor for Machine Learning-Based Prediction of Creep Rupture Life"
Collaborator: Idaho National Laboratory

Developed a temperature-dependent, geometry-corrected stress descriptor as an input feature for machine learning models that predict the creep rupture life of nuclear structural materials. Manuscript under review at *Journal of Nuclear Materials*.

Directed Research (First Author): "Domain-Adaptation Machine Learning Approach for Correlation of Full-Sized Charpy Properties from Sub-Sized Tests"
Collaborator: Idaho National Laboratory

Developed a domain-adaptation machine learning approach that correlates full-sized Charpy impact properties with measurements from sub-sized specimens. Manuscript under review.

Directed Research: "A Knowledge-Informed Large Language Model Framework for U.S. Nuclear Power Plant Shutdown Initiating Event Classification"
Collaborator: Idaho National Laboratory

Co-developed a two-stage hybrid framework for classifying shutdown initiating events (SDIEs) from U.S. nuclear power plant operating-experience reports to support shutdown probabilistic risk assessment. Addressed the lack of large labeled datasets, class imbalance, and label noise by designing a knowledge-informed prescreening stage based on 44 salient SDIE text patterns, followed by a BERT-based large language model that classifies SDIEs into four types. The prescreening stage excluded more than 97% of non-SDIEs, and the language model achieved 93.4% average classification accuracy. Findings were published in the *Journal of Risk and Reliability*. The team also developed a web-based tool implementing the framework, which was deployed to Idaho National Laboratory.

Directed Research: "Compilation of Creep Property Data for Nuclear Structural Materials"
Collaborator: Idaho National Laboratory

Contributed to compiling and curating a public, machine-learning-ready database of creep properties for nuclear structural materials. The dataset was released on the Materials Cloud Archive with open-source code and published in *Scientific Data*.

Undergraduate Capstone Project August 2022 – May 2023
Interdisciplinary Capstone Design Program, University of Idaho, Moscow, Idaho
Partner: NASA Ames Research Center

"Spear of Ares: Tube-Deployed Reentry Vehicle for Mars Weather Station"

Member of a seven-person interdisciplinary team that designed, built, and flight-tested a tube-deployed reentry vehicle (TDRV) prototype for potential use as a Mars weather station. The payload combined a spring-deployed

drag skirt, custom multilayer printed circuit boards, and an ESP32 flight computer running FreeRTOS with publish–subscribe inter-process communication, supported by a ground tracking GUI. A high-altitude balloon drop in February 2023 reached an apogee of 6,688 m MSL, with a measured terminal velocity of 12.7 m/s and a drag coefficient of 0.90. Presented at the University of Idaho Engineering Design EXPO 2023, where the team received the Best Booth Presentation award.

PUBLICATIONS

Manuscripts Under Review

Tao Wang, John W. Merickel, Yalei Tang, Fei Xu, Aleksandar Vakanski, and Rongjie Song. "A Temperature-Dependent Geometry-Corrected Stress Descriptor for Machine Learning-Based Prediction of Creep Rupture Life." Under review at *Journal of Nuclear Materials*.

Tao Wang, John W. Merickel, Yalei Tang, Fei Xu, Aleksandar Vakanski, and Rongjie Song. "Domain-Adaptation Machine Learning Approach for Correlation of Full-Sized Charpy Properties from Sub-Sized Tests." Under review.

Peer-Reviewed Journal Articles

Isshu Lee, John W. Merickel, Boopathy Kombaiah, Yalei Tang, **Tao Wang**, Fei Xu, Rongjie Song, and Aleksandar Vakanski. 2026. "Compilation of Creep Property Data for Nuclear Structural Materials." *Scientific Data* 13: 1262. https://doi.org/10.1038/s41597-026-07563-y

Min Xian, **Tao Wang**, Sai Zhang, Fei Xu, and Zhegang Ma. 2025. "A Knowledge-Informed Large Language Model Framework for U.S. Nuclear Power Plant Shutdown Initiating Event Classification for Probabilistic Risk Assessment." *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability* 239 (6): 1257–1264. https://doi.org/10.1177/1748006x251386900

DATASETS AND SOFTWARE

Creep Property Dataset for Nuclear Structural Materials. Materials Cloud Archive. https://doi.org/10.24435/materialscloud:99-xr

Web-based shutdown initiating event classification tool implementing the knowledge-informed LLM framework; deployed to Idaho National Laboratory.

HONORS & AWARDS

Best Booth Presentation, Engineering Design EXPO, University of Idaho	May 2023
Dean's List, University of Idaho	August 2022 – May 2023
Invitation to Idaho Scholarship, University of Idaho	August 2022 – May 2023

SKILLS

Research & Analysis:
Machine learning, large language models, natural language processing, text classification, transformer fine-tuning (BERT), domain adaptation, knowledge-informed feature engineering, learning from imbalanced and noisy labels, scientific data curation

Domain Knowledge:
Nuclear power plant safety, probabilistic risk assessment (PRA), nuclear structural materials, Charpy impact testing, creep and creep rupture behavior